A COMPOSITIONAL ACCOUNT OF MOTIFS, MECHANISMS, AND DYNAMICS IN BIOCHEMICAL REGULATORY NETWORKS.

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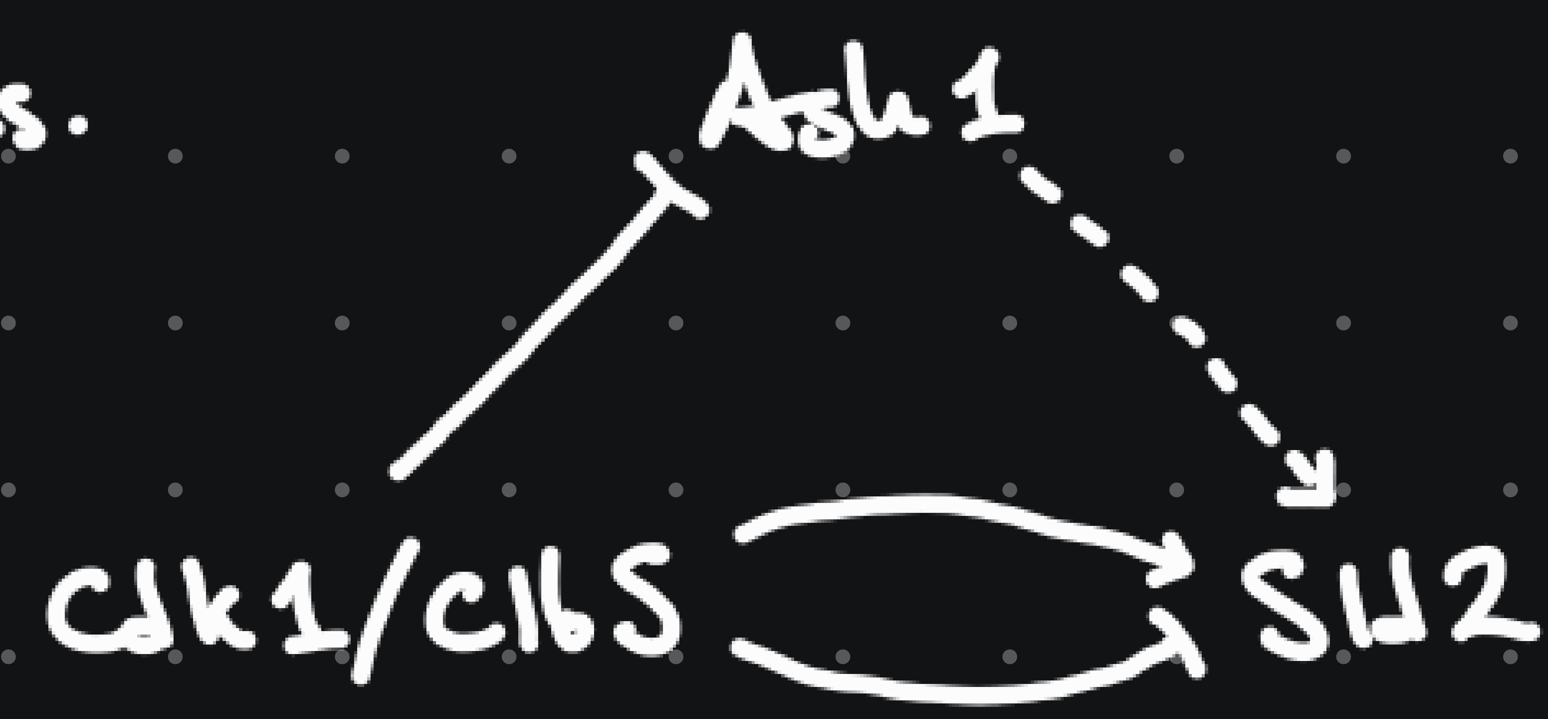
REGULATORY NETWORKS AND MOTIFS

A regulatory network conveys interactions between molecules.



- ★ Abundant in the literature.
- ★ Idealized and simplified model.
- ★ Imprecise treatment.

A motif is a simple recurring functionally meaningful pattern of regulatory networks.

Goal: Rigorous working definitions encoding functional interactions.

THE PRINCIPLE OF COMPOSITIONALITY

The meaning of a complex expression is determined by the combination of its constituents.

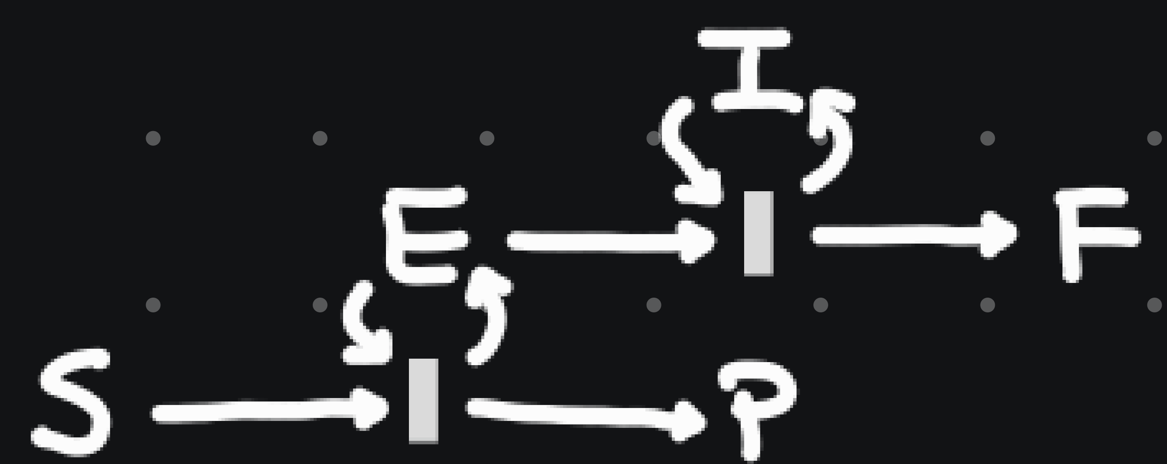
Definition: A <u>regulatory network</u> is a signed graph $\mathbb{X}$.

An <u>instance</u> of motif A in $\mathbb{X}$ is a monic signed functor $\mathrm{Path}(A) \rightarrowtail \mathrm{Path}(\mathbb{X})$.

Example: Unifying incoherent feedforward loops: $\{x \rightrightarrows y\} \rightarrowtail \{x \to y \dashv z\}$.

We model mechanics using Petri net with signed links.

Example: $I \dashv P$ ✗



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DYNAMICS AND LOTKA-VOLTERRA

We model dynamics using Lotka-Volterra, a generic model for entities that "regulate" each other.

★ Inspired but completely different from the Law of mass action on Petri nets.

Theorem: There exist functors giving the vector fields of Lotka-Volterra:

$$LV : \mathsf{FinGraph} \longrightarrow \mathsf{Para}(\mathsf{Dynam}) \text{ and } LV : \mathsf{FinSgnGraph} \longrightarrow \mathsf{Para}(\mathsf{Dynam}_+).$$

Example:

$$\begin{tikzcd}
\mathsf{FinSgnGraph} \arrow[d, "LV"] & A \quad B \arrow[r, hook, "\iota"] \arrow[d, dashed, "LV"] & A \underset{\dashv}{\overset{\to}{\rightleftarrows}} B \arrow[d, dashed, "LV"] \\
\mathsf{Para}(\mathsf{Dynam}_+) & \begin{cases} v_A(x;\rho) = -\rho_A x_A \\ v_B(x;\rho) = -\rho_B x_B \end{cases} \arrow[r, "LV(\iota)"] & \begin{cases} v_A(x;\rho,\beta) = -\rho_A x_A - \beta_{\dashv} x_A x_B \\ v_B(x;\rho,\beta) = -\rho_B x_B + \beta_{\to} x_A x_B \end{cases}
\end{tikzcd}$$

Thank you!